

WASP-12b

R-1173

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_161207 · created 2026-07-31T03:33:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457729.75726 +/- 0.00033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.103 +/- 0.044
Transit depth (Rp/Rs)^2	1.07 +/- 0.91 [%]
Orbital Inclination (inc)	86.76 +/- 2.82
Airmass coefficient 1 (a1)	0.26 +/- 0.15
Airmass coefficient 2 (a2)	1.32 +/- 0.65
Scatter in the residuals of the lightcurve fit is	66.86 %
Best Comparison Star	None
Optimal Aperture	3.72
Optimal Annulus	6.06
Transit Duration (day)	0.0889 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.103 ± 0.044 vs 0.1178 (deviation 0.27σ)
Duration fit vs published	0.0889 d vs 0.125 d (deviation 8.63σ)
Mid-time O–C	0.5 min
Depth SNR	1.8
Residual scatter	66.86 %

Light curve

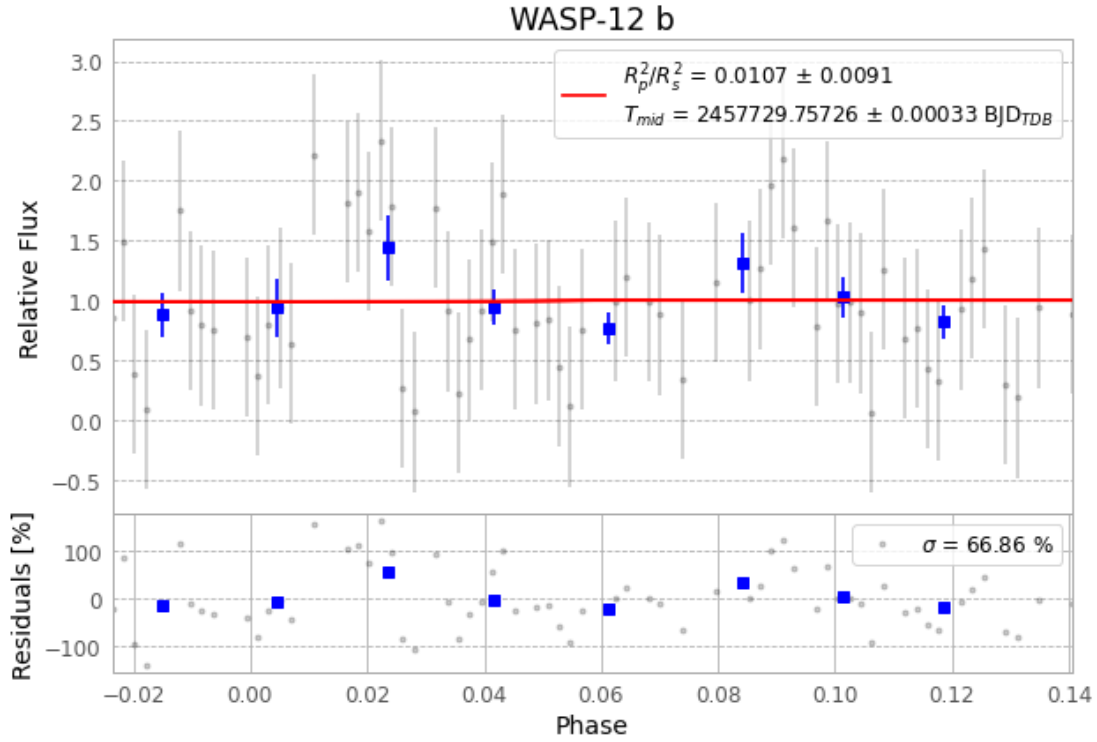


Figure 1 — FinalLightCurve_WASP-12 b_2016-12-06.png (SHA-256 beeb5ab829dd5c50...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aed89108f240e967...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), WASP-12161207052112.FITS → WASP-12161207094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161207.log (481e9ff27fb8...), FinalLightCurve_WASP-12 b_2016-12-06.png (beeb5ab829dd...), FinalLightCurve_WASP-12 b_2016-12-06.csv (3465980c2594...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 03:33Z.